

Digital Forensics and Incident Response (DFIR) Report

Report Date: Fri Oct 2 16:03:15 2026

Subject File: evidence02.pcapng

Analysis Scope: Packet Ranges 10–3711 and 7225–7442

Status: Final

1. Executive Summary

Analysis of `evidence02.pcapng` reveals two distinct periods of network activity involving internal host `10.0.0.210`. The first period is characterized by a sustained, high-volume outbound data transfer to an external, unidentified endpoint, consistent with data exfiltration or large-scale synchronization. The second period shows standard web activity involving Google services, interspersed with local network maintenance and evidence of network congestion/packet loss.

2. Key Findings

A. High-Volume Data Transfer (Exfiltration Indicator)

- **Activity:** A continuous, heavy outbound stream of encrypted SSLv2/TLS data was identified.
- **Endpoints:** Internal `10.0.0.210` → External `17.10.29.71`.
- **Behavior:** The internal host pushed large, 1460-byte segments to the destination. The remote server primarily responded with [ACK] packets, indicating it was receiving a continuous stream of data rather than serving content.
- **Assessment:** The directionality and volume are highly indicative of **unauthorized data exfiltration** or an unmanaged large-scale backup operation.

B. Web Services & DNS Activity

- **Activity:** Active TLSv1.2 sessions were observed interacting with Google infrastructure (`docs.google.com`, `ssl.gstatic.com`).
- **DNS Resolution:** Successful resolution of Google-owned IP addresses (IPv4 and IPv6) via standard DNS queries.
- **Session Termination:** Observed clean FIN, ACK handshakes and “Encrypted Alerts,” indicating standard protocol closures and/or error-driven terminations.

C. Network Anomalies & Infrastructure

- **Network Instability:** Evidence of packet loss and network congestion was detected (Packets 7346–7349), specifically “TCP Previous segment not captured,” “TCP Out-of-Order,” and “TCP Dup ACK” flags during high-volume transfers.
- **Local Network Traffic:**
 - **ARP:** Host 10.0.0.194 was actively attempting to locate the default gateway (10.0.0.1).
 - **Discovery:** Presence of SSDP/UPnP multicast traffic (239.255.255.250) and IPv6 Router Advertisements.
- **External Connection:** A connection to 128.119.245.12 (Port 80) was terminated by the remote host.

3. Indicators of Compromise (IoCs)

Indicator Type	Value	Context
External IP	17.132.29.71	Destination of high-volume outbound data transfer.
External IP	128.119.245.12	Remote host that terminated an HTTP session.
Network Behavior	Sustained Outbound TLS	Potential data exfiltration pattern.
Network Behavior	TCP Dup ACK / Out-of-Order	Evidence of congestion or capture-induced packet loss.

4. Conclusion & Recommendations

The primary concern is the sustained outbound data transfer from 10.0.0.210 to 17.132.29.71.

Recommended Actions: 1. **Host Investigation:** Perform a forensic audit of host 10.0.0.210 to identify the process initiating the connection to 17.132.29.71. 2. **Data Inventory:** Determine the nature of the data transmitted during the identified outbound burst. 3. **IP Reputation Check:** Verify the ownership and reputation of 17.132.29.71 and 128.119.245.12. 4. **Egress Filtering:** Review firewall logs to determine if other internal hosts are communicating with the identified external IPs.